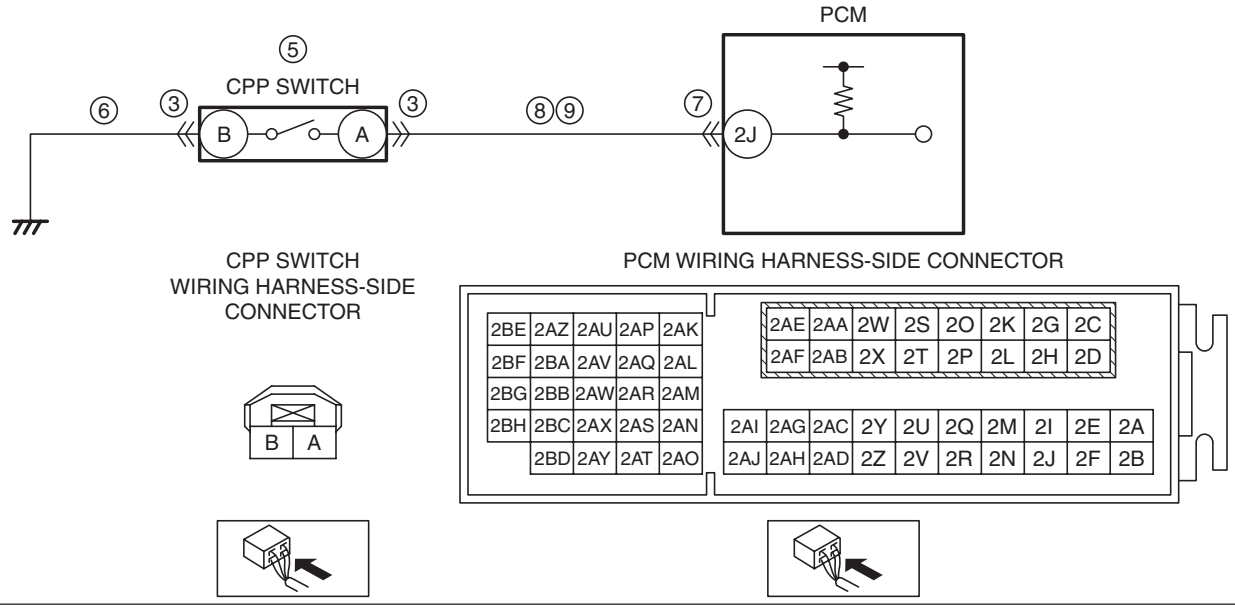
## DTC P0704:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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| DTC P0704:00 | CPP switch input circuit problem |
|---|---|
| DETECTION CONDITION | • The PCM monitors changes in input voltage from the CPP switch. If the PCM does not detect a voltage change while the vehicle runs with vehicle speed **above 30 km/h {19 mph}** and stops **8 times** alternately, the PCM determines that the CPP switch circuit has a malfunction.<br>**Diagnostic support note**<br>• This is a continuous monitor (CCM).<br>• The check engine light illuminates if the PCM detects the above malfunction condition in two consecutive drive cycles or in one drive cycle while the DTC for the same malfunction has been stored in the PCM.<br>• PENDING CODE is available if the PCM detects the above malfunction condition during first drive cycle.<br>• FREEZE FRAME DATA/Snapshot data is available.<br>• DTC is stored in the PCM memory. |
| FAIL-SAFE FUNCTION | Not applicable |
| POSSIBLE CAUSE | **Caution**<br>**• Inspect the CPP switch with it installed to the clutch pedal, otherwise the CPP switch may not operate normally after inspection. If the CPP switch is removed from the clutch pedal, replace the CPP switch with a new one.**<br><br>• CPP switch connector or terminals malfunction<br>• Foreign object lodged between clutch switch plunger and clutch pedal contact surface<br>• CPP switch malfunction<br>• Open circuit in wiring harness between CPP switch terminal B and body ground<br>• PCM connector or terminals malfunction<br>• Short to ground in wiring harness between CPP switch terminal A and PCM terminal 2J<br>• Open circuit in wiring harness between CPP switch terminal A and PCM terminal 2J<br>• Driver continues to depress clutch pedal causing mistaken detection by PCM<br>• PCM malfunction |




### Diagnostic Procedure

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION**<br><br>**Note**<br>• Recording can be facilitated using the screen capture function of the PC.<br><br>• Record the FREEZE FRAME DATA/snapshot data on the repair order. | — | Go to the next step. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 2 | **VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |
| 3 | **INSPECT CPP SWITCH CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the CPP switch connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 11. |
| | | No | Go to the next step. |
| 4 | **VERIFY IF MALFUNCTION OCCURS DUE TO FOREIGN OBJECT LODGED BETWEEN CLUTCH SWITCH PLUNGER AND CLUTCH PEDAL CONTACT SURFACE**<br>• Visually verify the area between the clutch switch plunger and clutch pedal contact surface.<br>• Is there a lodged foreign object? | Yes | Remove the foreign object, then go to the next step. |
| | | No | Go to the next step. |
| 5 | **INSPECT CPP SWITCH**<br>• Inspect the CPP switch. (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Is there any malfunction? | Yes | Replace the CPP switch, then go to Step 11.<br>(See CLUTCH PEDAL POSITION SWITCH REMOVAL/ INSTALLATION [F65M-R].)<br>(See CLUTCH PEDAL POSITION SWITCH REMOVAL/ INSTALLATION [F66M-R].) |
| | | No | Go to the next step. |
| 6 | **INSPECT CPP SWITCH GROUND CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the CPP switch connector is disconnected.<br>• Inspect for continuity between CPP switch terminal B (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal B and body ground.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Inspect for the following:<br>— Open circuit between CPP switch and body ground<br>— Loose or lifting ground point<br>• Repair or replace the malfunctioning part.<br>Go to Step 11. |
| 7 | **INSPECT PCM CONNECTOR CONDITION**<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 11. |
| | | No | Go to the next step. |
| 8 | **INSPECT CPP SWITCH SIGNAL CIRCUIT FOR SHORT TO GROUND**<br>• Verify that the CPP switch and PCM connectors are disconnected.<br>• Inspect for continuity between CPP switch terminal A (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal A and PCM terminal 2J.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to ground.<br>Go to Step 11. |
| | | No | Go to the next step. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 9 | **INSPECT CPP SWITCH SIGNAL CIRCUIT FOR OPEN CIRCUIT**<br>• Verify that the CPP switch and PCM connectors are disconnected.<br>• Inspect for continuity between CPP switch terminal A (wiring harness-side) and PCM terminal 2J (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal A and PCM terminal 2J.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 11. |
| 10 | **VERIFY THAT THERE IS NO PROBLEM WITH CUSTOMER CLUTCH PEDAL OPERATION**<br>• Is the clutch pedal depressed after operating the shift lever during a traffic jam? | Yes | Explain to the customer that it is necessary to remove the foot from the clutch pedal completely after operating the shift lever.<br>Go to the next step. |
| | | No | Go to the next step. |
| 11 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>**Caution**<br>**• While performing this step, always operate the vehicle in a safe and lawful manner.**<br>**• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.**<br>• Operate the clutch pedal while driving the vehicle **above 30 km/h {19 mph}** and stopping **8 times** alternately.<br>• Perform the Pending Trouble Code Access Procedure.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Is the PENDING CODE for this DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 12 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) |
| | | No | DTC troubleshooting completed. |